Dijkstra's Algorithm

shortest path <u>one</u> node to every other node.

1. Pick starting node A
2. Initialize tables
3. calculate distance unvisited neighbors of current node, update table if there is a shorter distance
4. Mark node as visited, choose new current unvisited node minimum table

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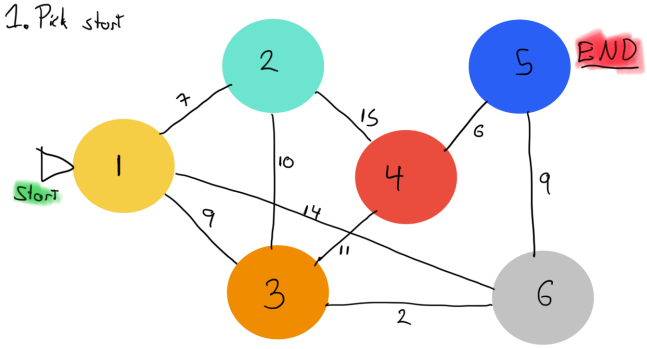
Example Assignment

1. Pick start



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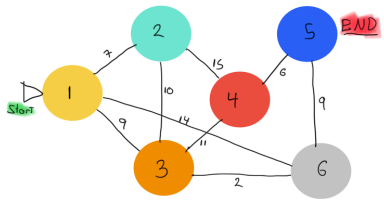
2. Initialize tables



Nodes	Shortest Distance	Previous node
1	0	
2	∞	
3	∞	
4	∞	
5	∞	
6	∞	

Visited [ ] Unvisited [1, 2, 3, 4, 5, 6]

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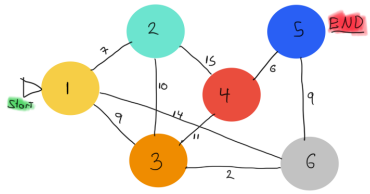
3. Calculate distance unvisited neighbors, update if shorter



Nodes	Shortest Distance	Previous node
1	0	
2	7	1
3	9	1
4	∞	
5	∞	
6	14	1

Visited [ ] Unvisited [1, 2, 3, 4, 5, 6]

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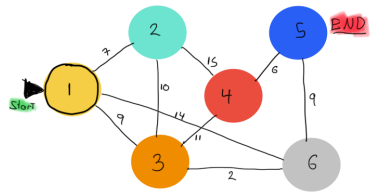
4. Mark node 1 as visited, choose new current unvisited node 2 min distance



Nodes	Shortest Distance	Previous node
1	0	
2	7	1
3	9	1
4	∞	
5	∞	
6	14	1

Visited [1] Unvisited [2, 3, 4, 5, 6]

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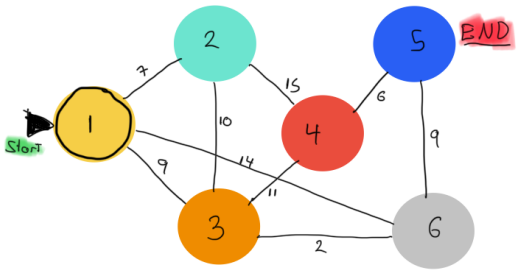
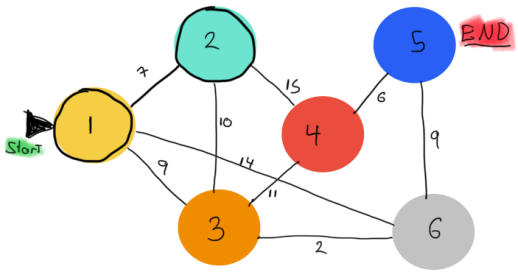
③ Dist neighbors, update



Nodes	Shortest Distance	Previous node
1	0	
2	7	1
3	9	1
4	22	2
5	∞	
6	14	1

Visited [1] Unvisited [2, 3, 4, 5, 6]

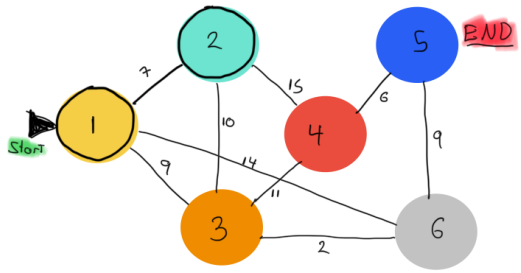
④ Mark 2 as visited, choose new current node 3



Nodes	Shortest Distance	Previous node
1	0	
2	7	1
3	9	1
4	22	2
5	∞	
6	14	1

Visited [1, 2] Unvisited [3, 4, 5, 6]

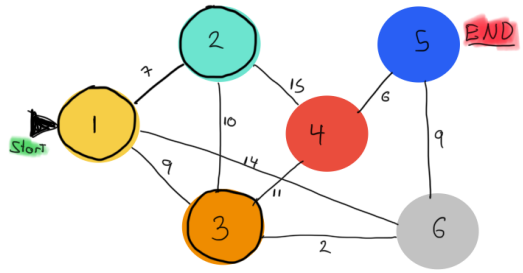
③ Dist neigbors, update



Nodes	Shortest Distance	Previous node
1	0	
2	7	1
3	9	1
4	20	3
5	∞	
6	11	3

Visited [1, 2] Unvisited [3, 4, 5, 6]

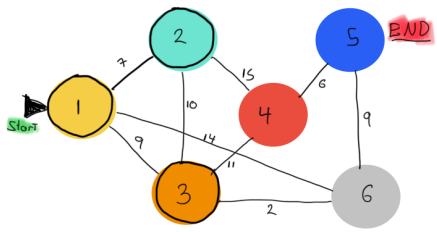
④ Visit 3, choose new current 6



Nodes	Shortest Distance	Previous node
1	0	
2	7	1
3	9	1
4	20	3
5	∞	
6	11	3

Visited [1, 2, 3] Unvisited [4, 5, 6]

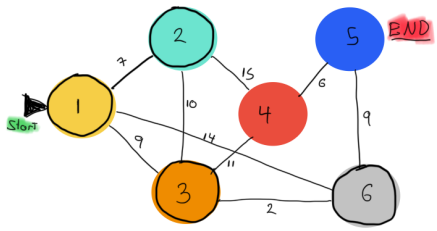
③ Dist neighbors, update



Nodes	shortest Distance	Previous node
1	0	
2	7	1
3	9	1
4	20	3
5	20	6
6	11	3

Visited [1, 2, 3] Unvisited [4, 5, 6]

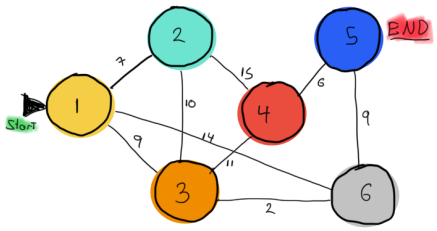
④ Mark 6 visited, choose 5 as next



Nodes	shortest Distance	Previous node
1	0	
2	7	1
3	9	1
4	20	3
5	20	6
6	11	3

Visited [1, 2, 3, 6] Unvisited [4, 5]

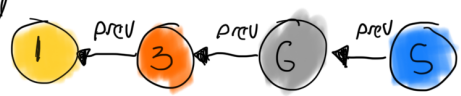
Dist neighbors, <u>no update</u>, Mark as visited 4 and 5



Nodes	shortest Distance	Previous node
1	0	
2	7	1
3	9	1
4	20	3
5	20	6
6	11	3

Visited [1, 2, 3, 6, 4, 5] Unvisited [ ]

From the table reconstruct shortest path 1 → 5

Print shortest distance = 20

Shortest Path